Nontrivial Multiples Preserving the Nonzero Digit Product and an Exact Finite Formula for OEIS A087304

Qichao Wang
Hebei University of Technology
Daoyu Dong
University of Electronic Science and Technology of China

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Abstract

Every positive integer has infinitely many multiples, other than shifts of its own digit string by trailing zeros, with the same product of nonzero digits. A repunit construction proves the nonvanishing conjecture recorded in OEIS A087304, in every integer base. For the decimal sequence we give an exact finite digit recurrence for its minimum. Its bound $n\tau(P)(d+2)$ depends only on the index n, its nonzero digit product P, and its digit length d.

1 A constructive existence proof

Let $P_b^*(n)$ be the product of the nonzero base-b digits of n. The empty product is 1. OEIS A087304 [1] asks for the smallest multiple M of n satisfying

$$P_{10}^*(M) = P_{10}^*(n), \qquad M \notin \{n10^j : j \geq 0\}.$$

Its entry records the conjecture that this minimum is always positive. The related A340204 [2], which includes zeros in its digit product, already records the usual repunit construction for numbers coprime to 10. We use that elementary idea after separating the base-supported part of the modulus.

Lemma 1. *If* $\gcd(m, b) = 1$*, there is a* $t \in \{1, \ldots, m\}$ *such that* $m \mid R_{b,t}$*, where* $R_{b,t} = 1 + b + \cdots + b^{t-1}$*.*

Proof. Consider the m residues of $R_{b,1}, \ldots, R_{b,m}$. If none is zero, two coincide. Their difference is $b^j R_{b,t}$ for some $1 \leq t < m$, and b is invertible modulo m. This also includes $m = 1$. □

Theorem 1. *Fix* $b \geq 2$ *and* $n \geq 1$*. Factor* $n = qm$ *with* $\gcd(m, b) = 1$ *and every prime divisor of* q *dividing* b*. Choose* $s \geq 0$ *with* $q \mid b^s$*, and choose* $t \leq m$ *as in Lemma 1. Then, for every* $j \geq 1$*,*

$$M_j = (nb^{jt} + R_{b,jt})b^s \tag{1}$$

is divisible by n*, preserves* $P_b^*(n)$*, and is not* nb^h *for any* $h \geq 0$*. These witnesses are unbounded. In particular, A087304 has no zero terms, and a witness has at most* $d_b(n) + s + m$ *digits.*

Proof. The identity $R_{b,jt} = R_{b,t}(1 + b^t + \cdots + b^{(j-1)t})$ shows that $m \mid R_{b,jt}$. Hence $n \mid b^s R_{b,jt}$ and $n \mid M_j$. The representation of M_j is precisely the digits of n, followed by jt ones and s zeros, so its nonzero digit product is unchanged. Finally,

$$b^{jt+s} < M_j/n = b^s(b^{jt} + R_{b,jt}/n) < b^{jt+s+1},$$

using $0 < R_{b,jt} < b^{jt}$ and $2 \leq b$. No integral power of b lies strictly between these consecutive powers. Taking $j = 1$ gives the digit bound, and letting j increase gives infinitely many witnesses. □

2 An exact finite formula for the least multiple

We now work in base ten. Put $P = P^*_{10}(n)$ and $d = d_{10}(n)$, and write the digits of n as $c_1 \cdots c_d$. Let $\mathcal{D}(P)$ be the positive divisors of P, and put

$$\mathcal{S} = \{0, \ldots, n-1\} \times \mathcal{D}(P) \times \{0, \ldots, d+1\}, \qquad B = |\mathcal{S}| = n\tau(P)(d+2).$$

A state (r, q, h) records the current remainder, the unused factor of the target product, and a state recognizing the forbidden words $c_1 \cdots c_d 0^*$. Here $h < d$ means that exactly h initial digits have matched, $h = d$ means a complete match followed only by zeros, and $h = d+1$ means a permanent mismatch. Define

$$H(h, a) = \begin{cases} h+1, & h < d \text{ and } a = c_{h+1}, \\ d+1, & h < d \text{ and } a \neq c_{h+1}, \\ d, & h = d \text{ and } a = 0, \\ d+1, & \text{otherwise.} \end{cases}$$

For $a \in \{0, \ldots, 9\}$ let $w(a) = \max(1, a)$. A digit is admissible when $w(a) \mid q$, and the transition is

$$T_a(r, q, h) = ((10r + a) \bmod n,\ q/w(a),\ H(h, a)). \tag{2}$$

Introduce a separate initial state $*$, with transitions corresponding to the admissible digits $1, \ldots, 9$ applied to $(0, P, 0)$. Thus leading zeros and the empty word are excluded. The accepting states are

$$\mathcal{F} = \{(0, 1, h) : h \neq d\}.$$

Define $D_0(*) = 0$, all other $D_0 = +\infty$, and for every state v set

$$D_{\ell+1}(v) = \min_{u \xrightarrow{a} v} \big(10 D_\ell(u) + a\big), \qquad \min \varnothing = +\infty. \tag{3}$$

Theorem 2 (All-index minimum formula)**.** *With the explicit states and transitions above,*

$$\mathrm{A087304}(n) = \min_{1 \le \ell \le B} \ \min_{v \in \mathcal{F}} D_\ell(v). \tag{4}$$

In particular its least witness has at most B decimal digits.

Proof. Induction on the word length proves the interpretations of r, q, h. Consequently the accepted words are exactly the eligible multiples of n. Their positivity and divisibility exclude values below n, and the forbidden-word state excludes n and all its decimal shifts. Theorem 1 ensures at least one accepted word exists.

On a shortest accepting path, no state can repeat: removing the segment between two occurrences leaves an accepting path. If the repetition is after the initial state, its first nonzero digit is retained; the separate initial state has no incoming edges. There are $B + 1$ states including $*$, so such a path has at most B edges. A positive decimal integer with fewer digits is smaller than one with more digits. For each fixed length, (3) gives the smallest label reaching each state: appending the same digit preserves the ordering of prefix values. Thus (4) gives the global least eligible integer. □

Formula (4) is a finite minimum with a fixed number of digit recurrence steps; it does not require scanning successive multiples. The accompanying program evaluates it by breadth-first search with digits ordered increasingly. State merging preserves the shortest, lexicographically first prefix. Independent direct enumeration verifies the initial published minima. The already recorded identity $a(10n) = 10a(n)$, credited to Chai Wah Wu (2017) in [1], also follows by deleting or appending one trailing zero in an eligible multiple.

References

[1] A. Murthy, OEIS A087304, https://oeis.org/A087304, accessed October 26, 2026.

[2] B. Schott, OEIS A340204, https://oeis.org/A340204, accessed October 26, 2026.